

## Internship Details

AlgoUniversity (backed by Y-Combinator)

June 2025 - July 2025

**Github Link:** [https://github.com/Diven2510/ATF\\_Project](https://github.com/Diven2510/ATF_Project)

**Demo Video :** <https://www.loom.com/share/11e49d2f977340d08379d60f4399f6ad?sid=6c4cbcec-0b7a-40fc-8ec7-2b3e93ab0495>

**Live Website:** [algotesters.live](http://algotesters.live)

- Developed a full-stack web application for competitive programming, enabling users to solve coding problems, participate in contests, and view real-time leaderboards.
- Built the frontend using React, Tailwind CSS, and Monaco Editor for a modern, responsive UI with syntax-highlighted code editing and dynamic dark/light mode.
- Engineered a Node.js/Express backend with MongoDB (Mongoose) for secure user authentication (JWT), problem management, and persistent code submissions.
- Designed and deployed a Dockerized code compiler microservice (C++, Python, Java) using child process isolation for safe, real-time code execution and test case evaluation.
- Integrated Google Gemini AI API to auto-generate problem metadata, hints, and code explanations, enhancing user learning and admin productivity.
- Implemented admin dashboard for problem creation, editing, and deletion, with AI-powered test case and example generation.
- Deployed the platform on AWS EC2 with Nginx reverse proxy and SSL, ensuring secure, scalable, and reliable access for users.

## **Other Projects:-**

### **Github timeline Email Notification System**

I have developed a PHP-based GitHub Timeline Email Notification System that allows users to subscribe using their email address to receive periodic updates of their GitHub activity. When a user enters their email on the web interface (index.php), a 6-digit verification code is sent to their inbox using SMTP via PHPMailer. Upon successful verification, the email is stored in registered\_emails.txt. We've implemented a CRON job mechanism (cron.php), set up through the setup\_cron.sh script, which runs every 5 minutes and sends a formatted HTML email containing the latest GitHub timeline updates to all registered users. Additionally, each email includes an unsubscribe link that allows users to opt out. Clicking this link triggers a confirmation flow, where a new code is emailed, and upon correct entry, the email is removed from the system. All sensitive credentials like email and password are securely managed using a .env.ini file, which is excluded from version control through .gitignore. The entire functionality resides within the /src directory, and the application has been tested successfully using XAMPP on Windows.

### **Thrust Stand**

A thrust stand is a device used to measure the thrust produced by propulsion systems such as electric motors, jet engines, or rocket engines. By implementing a thrust stand, I have created a setup that accurately captures the force generated during engine or motor operation. The system typically includes a load cell or force sensor mounted on a stable frame, along with a data acquisition system to monitor and record thrust values in real time. This implementation is crucial for performance testing, calibration, and optimization of propulsion systems, ensuring precise and reliable measurements for research or engineering applications.

### **Personal Finance Management System**

Personal Finance Management System is a Java-based desktop application developed using Java Swing that allows users to manage their income and expenses efficiently. The system provides features to record transactions, categorize them, and store data securely using a MySQL database via JDBC. It offers real-time financial insights through visualizations like bar and pie charts using the XChart library. The UI is user-friendly, with modules for input, analysis, and reporting. Users can also generate PDF reports of transactions, track monthly overviews, and view remaining balances. This project helps promote financial awareness and simplifies daily expense tracking in a structured and visual manner.